AI-ROBO

INDUSTRIAL SMART AUTOMATION SOLUTION

AI Collaborative Robot System Hardware Installation Guidelines

This document was produced based on the hardware technical guidelines in the Guide tab of the official AI-ROBO website.

Document Type	Technical Installation Manual
Source	AI-ROBO Engineering Solution Portal
Version	v1.1 (2026 Edition)
Safety Cert.	ISO 10218-1 / ISO/TS 15066 Compliant

1. AI-ROBO Hardware Solution Overview

AI-ROBO smart collaborative robots are designed for safe coexistence with humans and high productivity. In order to operate flexibly in the same space as workers without a safety fence, the initial physical fixation and power system connection must be perfectly implemented.

> ⚠️ **Read: Safety Notice**
>
> Although AI-ROBO is equipped with 24/7 real-time monitoring and autonomous collision detection, poor physical fixation or electrical noise may cause sensor malfunction. Disconnect the main power during all assembly and wiring stages.

1.1 Required Infrastructure Specifications

Category	Requirements	Importance
Flatness	Slope under 0.3°, steel frame or concrete floor without vibration	Highest
Power Spec	AC 200V ~ 240V, 50/60Hz (Single phase), voltage fluctuation within ±10%	Required
Grounding	Class 3 grounding (Dedicated grounding under 100Ω recommended)	Required
Pneumatic	Purified compressed air 0.5 ~ 0.7 MPa (When using pneumatic gripper)	Optional

2. Step 1: Robot Base Mechanical Fixation

2.1 Positioning and Leveling

- Review the layout to ensure no physical interference within the robot's maximum workspace.
- Place a precision digital level at the center of the base and adjust levelers within 0.3°.

2.2 High-Tensile Bolt Tightening Procedure

1. Apply medium-strength threadlocker (Loctite 243) to high-tensile hex bolts matching the AI-ROBO base specification.

2. Tighten bolts gradually in a **diagonal pattern sequence** to ensure even force distribution.
3. Complete final tightening using a torque wrench until the specified torque value for each model is reached.

3. Step 2: Power Infrastructure & Signal Interface

3.1 Main Power and Controller Wiring

- Design the power wiring to pass through a dedicated Earth Leakage Circuit Breaker (ELB) at the input stage.
- Maintain a distance of at least 10cm between power and communication lines to prevent induction noise.

3.2 Harness Connector Coupling

When connecting the heavy-duty connector linking the robot body and controller, push it vertically to avoid bending internal pins, then lock the external dual locking levers completely until it clicks.

4. Step 3: End-Effector & Line Dress Guide

This process involves attaching appropriate tools (grippers, vision cameras, etc.) to the mechanical flange at the end of the AI-ROBO arm.

4.1 Dresspack Optimization Method

To prevent cables and air hoses from pulling tightly or twisting during joint rotation, perform a virtual **maximum workspace test**, secure sufficient margin, and clamp them along the arm.

5. Comprehensive Hardware Verification List

Item	Detailed Inspection Criteria	Result
Mechanical Fixation	Are the base fixing bolts completely finished with the specified torque wrench value?	[ ] Pass / [ ] Fail
Power System	Is the supply voltage normal and is the Class 3 grounding completed?	[ ] Pass / [ ] Fail
Cabling	Are the heavy-duty connector levers between the controller and robot completely locked?	[ ] Pass / [ ] Fail
Interference	Is there no interference with surrounding structures or cables during robot/tool operation?	[ ] Pass / [ ] Fail